Sana'a University
Faculty of Computer and Information Technology
Master in Cyber Security

MASTER RESEARCH PROPOSAL

Decentralized Dynamic DNS as a Threat to ISP-Level Network Control: Detection, Threat Quantification, and Mitigation

Design, Implementation, and Evaluation

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Proposal positioning

This proposal is organized around the supervisor's chapter-based research methodology: Chapter 1 establishes the problem and research foundation, Chapter 2 synthesizes the literature and gap, and Chapter 3 defines the quantitative experimental design, baselines, metrics, and evidence plan.

ABSTRACT

Internet Service Providers (ISPs) have historically relied on their position in the DNS resolution path to exercise network-level control: content filtering, parental-control enforcement, lawful interception, malware command-and-control (C2) blocklisting, and abuse-response tracking of dynamically assigned addresses. The emergence of decentralized Dynamic DNS (DDNS) mechanisms, blockchain-anchored naming systems such as Namecoin, Handshake, and the Ethereum Name Service (ENS), together with encrypted resolution transports such as DNS-over-HTTPS (DoH), DNS-over-TLS (DoT), and DNS-over-QUIC (DoQ), removes the ISP from the naming and resolution path entirely. Domain records anchored on a public ledger and updated peer-to-peer are neither centrally registered nor visible in plaintext to the access network, which erodes the technical basis for ISP-level filtering, monitoring, and abuse mitigation.

This research investigates decentralized DDNS as a structural threat to ISP-level network control. The study characterizes the threat model, in which both privacy-seeking end users and malicious actors, including botnet and malware operators seeking censorship-resistant and takedown-resistant C2 infrastructure, adopt decentralized naming and encrypted resolution to bypass conventional DNS-based governance. The study then designs, implements, and evaluates a network-edge detection and control-recovery framework that combines encrypted-traffic fingerprinting and machine-learning classification with a quantified "control-loss index" to measure how much ISP-level visibility and filtering capability is lost under increasing decentralized-DDNS adoption, and how much can be recovered through targeted, low-overhead mitigation. The framework is evaluated in a controlled network testbed against traffic mixes representative of legitimate blockchain-DNS use, encrypted DNS tunneling, and blockchain-anchored malware C2, measuring detection accuracy, false-positive impact on legitimate encrypted traffic, latency and throughput overhead, and blocklist-maintenance cost. The expected contribution is a reproducible, quantitative account of how decentralized DDNS erodes ISP-level control, together with a practical, privacy-conscious detection and mitigation framework that ISPs and network defenders can evaluate against their own operational constraints.

Keywords: Decentralized DNS; Dynamic DNS; ISP-level control; blockchain DNS; Handshake; Namecoin; ENS; DNS-over-HTTPS; encrypted traffic classification; malware command-and-control resilience; Protective DNS.

Contents

Abstract **2**
Keywords 2
Proposal At-a-Glance **6**
Alignment with the Supervisor's Chapter Guides **7**
Chapter 1: Introduction and Research Foundation **8**
1.1 Background and Context 8
1.2 Research Problem 8
1.3 Scientific Gap 8
1.4 Purpose Statement / Aim 9
1.5 Research Questions 9
1.6 Research Objectives 10
1.7 Research Hypothesis 10
1.8 Significance of the Study 10
1.9 Scope and Delimitations of the Study 10
1.10 Limitations of the Study 11
1.11 Definition of Key Terms 11
1.12 Organization of the Thesis 11
Chapter 2: Literature Review and Theoretical Background **13**
2.1 Introduction 13
2.2 Literature Search Logic and Screening Criteria 13
2.3 Traditional DNS and the Basis of ISP-Level Control 13
2.4 Blockchain-Anchored and Peer-to-Peer Naming Systems 14
2.5 Encrypted DNS Transports and the Erosion of Visibility 14
2.6 Abuse of Decentralized and Encrypted Naming by Malicious Actors 14
2.7 Detection and Classification of Encrypted/Decentralized DNS Traffic 14
2.8 Standards and Governance Blueprints 14
2.9 Comparative Review of Previous Studies and Technical References 15
2.10 Synthesized Research Gap 16
2.11 Conceptual Framework of the Proposed Study 17
Chapter 3: Research Methodology and Experimental Design **18**
3.1 Research Design 18
3.2 Quantitative Nature of the Study 18
3.3 Proposed Artifact and System Architecture 18
3.4 Technical Design of the Detection and Mitigation Scheme 19
3.5 Threat Model and Assumptions 19
3.6 Unit of Analysis and Observation 20
3.7 Baseline Approaches and Planned Comparison 20
3.8 Variables and Metrics 21
3.9 Research Question, Objective, and Metric Mapping 21
3.10 Experimental Environment 22
3.11 Traffic and Threat Scenarios 22
3.12 Experimental Protocol 22
3.13 Data Collection and Evidence Logging 23
3.14 Data Analysis Plan 23
3.15 Statistical Treatment and Repetition Rules 24
3.16 Validity, Reliability, and Reproducibility 25

3.17 Ethical Considerations 25
3.18 Expected Results 25
3.19 Proposed Timeline 26
Proposal Readiness Checklist 27
Conclusion 28
References 29

List of Figures

1 Research gap and positioning of the proposed study 9
2 Proposed detection and control-recovery architecture 19
3 Threat model and control-erosion pathways 20
4 Proposed research methodology and evaluation pipeline 23
5 Evaluation scenario and comparative baseline framework 24

PROPOSAL AT-A-GLANCE

Research spine

The proposal is deliberately focused on one defensible research problem: quantifying, and partially recovering, the loss of ISP-level network control caused by decentralized Dynamic DNS and encrypted-resolution technologies. It avoids broad claims such as "breaking blockchain DNS" or "ending internet privacy" and instead defines a measurable artifact, comparison baselines, realistic traffic scenarios, technical metrics, and a reproducible evidence plan.

Element	Final Research Decision
Core problem	Decentralized DDNS mechanisms (blockchain-anchored naming plus encrypted resolution transports) remove the ISP from the DNS resolution path, eroding content filtering, lawful interception, malware-C2 blocking, and abuse-tracking capability that network operators traditionally rely upon.
Scientific gap	Blockchain-DNS research emphasizes system design and censorship resistance; encrypted-DNS research emphasizes enterprise visibility loss and detection; malware-defense research emphasizes C2 resilience. Few studies quantify the combined, network-operator-facing loss of control or evaluate a unified, low-overhead detection and mitigation framework across both decentralized naming and encrypted transport.
Proposed artifact	A network-edge detection pipeline combining encrypted-traffic feature extraction and machine-learning classification, a quantified "control-loss index," and a targeted, privacy-conscious mitigation policy layer.
Evaluation logic	Compare the proposed detection-and-mitigation framework against ISP status-quo practices (legacy DNS filtering, do-nothing, Protective-DNS-only) under identical traffic mixes of legitimate, privacy-seeking, and malicious decentralized-DDNS usage.
Primary metrics	Detection accuracy/F1, false-positive rate on legitimate encrypted traffic, control-loss index before/after mitigation, malicious-C2 block success rate, inline latency/throughput overhead, and blocklist-maintenance churn.
Boundaries	The study addresses network-edge, ISP-operator-facing detection and mitigation; it does not attempt to break blockchain consensus or cryptographic naming guarantees, and it treats legal, regulatory, and policy questions as context rather than as a primary object of empirical study.

ALIGNMENT WITH THE SUPERVISOR'S CHAPTER GUIDES

Guide Requirement	Where It Appears	How the Proposal Satisfies It
Problem statement	Chapter 1, Sections 1.2–1.3	Maps the technical mechanisms by which decentralized DDNS removes ISP visibility and the concrete absence of an integrated, quantified response.
Purpose statement	Chapter 1, Section 1.4	Formulates a quantitative, engineering-oriented purpose statement grounded in controlled experimentation.
Research questions	Chapter 1, Section 1.5	Details specific, answerable inquiries targeting threat characterization, detectability, and control-recovery trade-offs.
Objectives	Chapter 1, Section 1.6	Transforms each research question into an achievable engineering and evaluation task.
Significance	Chapter 1, Section 1.8	Demonstrates theoretical, practical, methodological, security, and governance value.
Scope and limitations	Chapter 1, Sections 1.9–1.10	Enforces clear boundaries on network layer, dataset scope, and what the study does and does not claim about privacy or legality.
Literature review	Chapter 2	Structures a component-based analysis across blockchain DNS, encrypted transports, malware abuse, and detection research, culminating in an empirical gap.
Methodology	Chapter 3	Outlines rigorous experimental conditions, independent/dependent variables, traffic-generation rules, and data-logging procedures.

CHAPTER 1: INTRODUCTION AND RESEARCH FOUNDATION

1.1 Background and Context

The Domain Name System (DNS) has historically placed Internet Service Providers (ISPs) directly in the resolution path between a subscriber and the rest of the internet. Because DNS queries were traditionally sent in plaintext to resolvers the ISP operates or observes, network operators could inspect, log, filter, or redirect lookups to enforce content policy, comply with lawful-interception orders, block known malware command-and-control (C2) domains, and correlate dynamically assigned IP addresses with abuse reports. This position has functioned as one of the few practical, scalable control points available to network operators.

Two converging technical trends now bypass that control point. First, blockchain-anchored and peer-to-peer naming systems such as Namecoin, Handshake, and the Ethereum Name Service (ENS) allow names to be registered, updated, and resolved without a centrally operated registry or resolver, replacing ICANN-style governance with cryptographic consensus. Second, encrypted DNS transports, DNS-over-HTTPS (DoH), DNS-over-TLS (DoT), and DNS-over-QUIC (DoQ), remove the plaintext visibility that made ISP-side inspection possible even for conventional, centrally registered domains. When decentralized naming and encrypted resolution are combined into decentralized Dynamic DNS (DDNS) services that keep a self-hosted or mobile endpoint continuously reachable under a blockchain-anchored name, the ISP loses both the registry-level and the transport-level visibility it previously relied on.

1.2 Research Problem

ISPs and network defenders face a compounding visibility problem. Individually, blockchain DNS and encrypted DNS transports are each the subject of active research; together, and specifically in the dynamic-update use case where a name must be re-published whenever an endpoint's address changes, they create a naming and resolution channel that is simultaneously outside central registry control and outside plaintext network observation. This channel is attractive to privacy-conscious users, but it is equally attractive to malware operators seeking takedown-resistant C2 infrastructure: a blockchain-anchored name cannot be seized or sinkholed by a registrar, and DoH/DoT/DoQ transport prevents signature-based blocking of the underlying domain string.

Current ISP practice is not equipped for this shift. Legacy DNS-based content filtering and Protective DNS deployments assume that the ISP's resolver, or at least plaintext traffic to a visible resolver, is on-path. Where encrypted DNS is used with a small number of well-known public resolvers, ISPs can still apply IP-based policy; but decentralized DDNS resolution can occur through locally run full nodes, browser extensions, or gateway services whose endpoints are neither exhaustively enumerable nor stable. The result is a widening, largely unquantified gap between the control ISPs believe they have and the control they can actually exercise.

1.3 Scientific Gap

The literature addresses the pieces of this problem separately. Blockchain-DNS research (Namecoin, Handshake, ENS, and comparative surveys) focuses on system design, decentralization guarantees, and censorship resistance from the perspective of the naming system's users and operators. Encrypted-DNS research documents the loss of enterprise and ISP visibility caused by DoH/DoT/DoQ and proposes machine-learning classifiers to detect encrypted DNS traffic, but largely in a generic “encrypted vs. unencrypted” framing rather than specifically targeting decentralized-DDNS resolution and update traffic. Malware-defense research documents that blockchain-based naming is already used for C2 resilience and discusses the resulting challenges for defenders, but does not provide an ISP-operator-facing detection and mitigation architecture with quantified control-loss and control-recovery metrics. This proposal occupies the resulting gap: a network-edge, quantitatively evaluated framework that measures how much ISP-level control is lost to decentralized DDNS and how much can be recovered through targeted detection and mitigation, without resorting to blanket encrypted-traffic blocking.

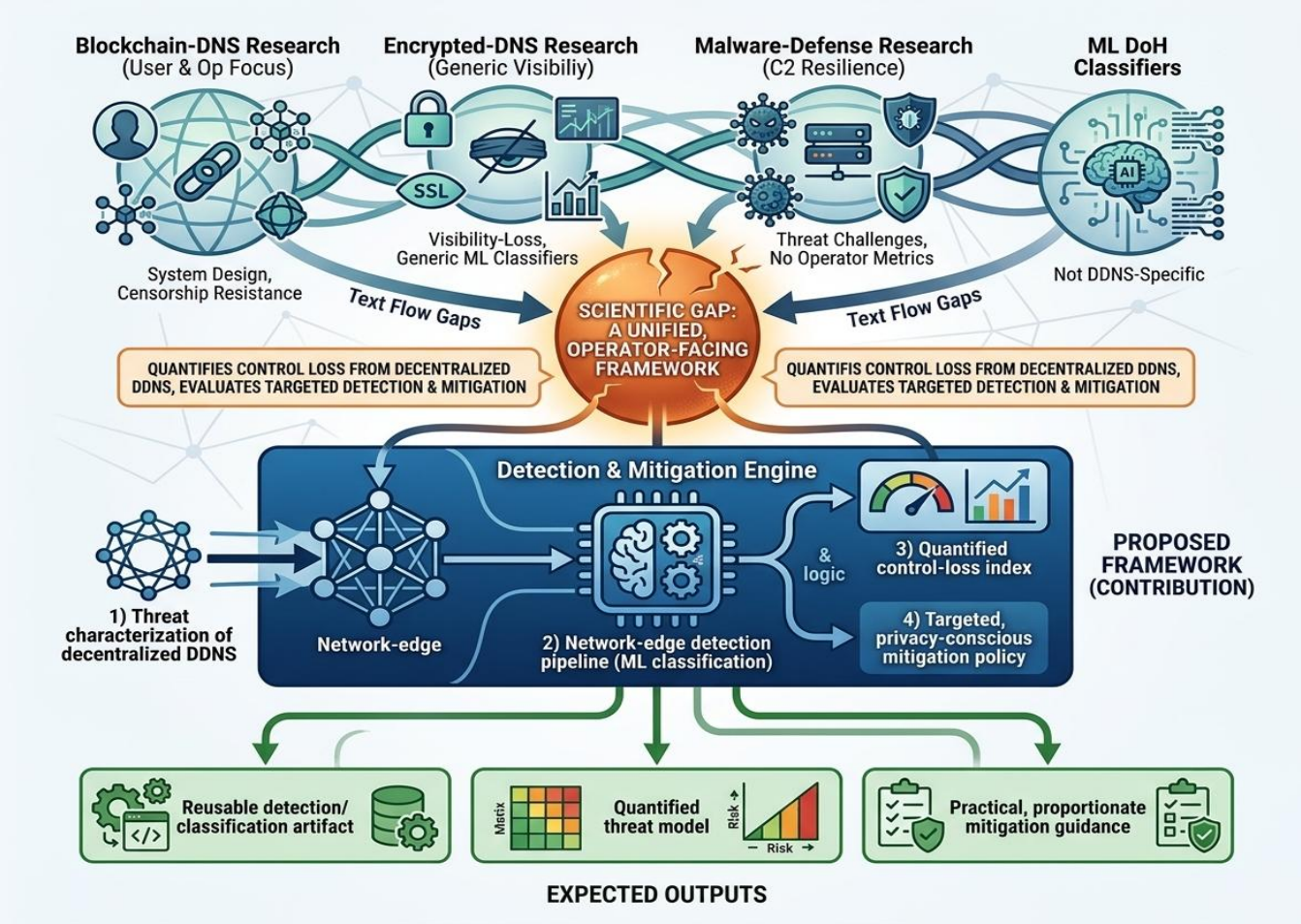


Figure 1: Research gap and positioning of the proposed study.

1.4 Purpose Statement / Aim

The purpose of this quantitative experimental design-science study is to characterize, detect, and partially mitigate the erosion of ISP-level network control caused by decentralized Dynamic DNS. The study designs, implements, and evaluates a network-edge detection and control-recovery framework, and compares it against current ISP practice in terms of detection accuracy, impact on legitimate privacy-seeking traffic, malicious-C2 blocking effectiveness, and operational overhead.

1.5 Research Questions

- RQ1. Which decentralized DDNS mechanisms and encrypted-transport combinations pose the greatest practical loss of ISP-level visibility and control, and why?
- RQ2. How does the combination of blockchain-anchored naming with encrypted resolution transports (DoH/DoT/DoQ) compare to each mechanism in isolation, in terms of the fraction of naming activity that remains observable to a network operator?
- RQ3. Can a network-edge machine-learning classifier reliably distinguish decentralized-DDNS resolution and update traffic from ordinary encrypted web traffic without excessive false positives?
- RQ4. To what extent can targeted detection and policy enforcement recover ISP-level control (content filtering, malicious-C2 blocking, abuse tracking) without materially degrading legitimate encrypted traffic or requiring blanket protocol blocking?

- RQ5. What performance costs, in latency, throughput, and blocklist-maintenance effort, does the proposed detection and mitigation framework impose under realistic ISP traffic volumes?

1.6 Research Objectives

1. To characterize the technical mechanisms by which decentralized DDNS and encrypted resolution transports

The objectives below have been restructured so that each of the five research objectives maps to exactly one research question, on a strict one-to-one basis. This replaces the earlier six-objective formulation, in which two objectives (literature synthesis and standards mapping) did not correspond to an individually evaluated research question. Table 1.6 makes the mapping explicit; the literature synthesis and the standards/governance alignment are retained as supporting activities within Chapters 2 and 3 rather than as separately numbered, separately evaluated objectives.

Objective	Research Question	Statement
Objective 1	RQ1	To characterize which decentralized-DDNS mechanisms and encrypted-transport combinations pose the greatest practical loss of ISP-level visibility and control, and to explain the technical basis for that loss.
Objective 2	RQ2	To quantify the fraction of naming/resolution activity that remains observable to a network operator when blockchain-anchored naming is combined with encrypted resolution transports (DoH/DoT/DoQ), compared with each mechanism used in isolation.
Objective 3	RQ3	To design, implement, and evaluate a network-edge machine-learning classifier that distinguishes decentralized-DDNS resolution and update traffic from ordinary encrypted web traffic, reporting detection accuracy and false-positive rate.
Objective 4	RQ4	To implement a targeted, proportionate mitigation-policy layer and evaluate the extent to which it recovers ISP-level control — content filtering, malicious-C2 blocking, and abuse tracking — without materially degrading legitimate encrypted traffic or requiring blanket protocol blocking.
Objective 5	RQ5	To measure the performance costs of the proposed detection-and-mitigation framework — inline latency, throughput impact, and blocklist-maintenance effort — under realistic ISP traffic volumes.

1.7 Research Hypothesis

A network-edge detection pipeline that combines encrypted-traffic fingerprinting with machine-learning classification, paired with a targeted and proportionate mitigation policy, will significantly reduce the measured control-loss index attributable to decentralized DDNS, and will substantially improve blocking of blockchain-anchored malicious C2 infrastructure, while keeping false-positive impact on legitimate encrypted traffic and inline performance overhead within operationally acceptable bounds.

1.8 Significance of the Study

Dimension	Significance
Theoretical significance	Connects blockchain-DNS design research, encrypted-transport research, and malware-defense research into a single, quantified threat model of ISP-level control erosion.

Dimension	Significance
Practical significance	Delivers an actionable detection pipeline and mitigation policy template that ISPs and network defenders can evaluate for content filtering, abuse response, and malicious-C2 blocking.
Methodological significance	Provides a reproducible, quantitative "control-loss index" methodology that other researchers or operators can apply to new decentralized-naming technologies as they emerge.
Security significance	Improves the ability to detect and block blockchain-anchored malware C2 infrastructure without resorting to disproportionate blanket blocking of encrypted DNS.
Governance significance	Informs the policy debate about encrypted DNS and decentralized naming by providing measured, rather than anecdotal, evidence of the resulting control trade-offs.

1.9 Scope and Delimitations of the Study

This study is conducted from the perspective of a network operator (ISP or large enterprise network) attempting to detect and respond to decentralized-DDNS traffic at the network edge.

Included in the Study	Excluded from the Study
Blockchain-anchored naming systems (Namecoin, Handshake, ENS-style resolution) and their traffic characteristics.	Attacks against blockchain consensus, cryptographic protocol weaknesses, or naming-system internals.
Encrypted DNS transports (DoH, DoT, DoQ) as used to resolve both centralized and decentralized names.	Legal, regulatory, and policy analysis of net neutrality, censorship law, or lawful-interception mandates, beyond contextual discussion.
Network-edge traffic fingerprinting and machine-learning classification of decentralized-DDNS activity.	End-user device compromise, malware analysis beyond its C2 naming/resolution behaviour, and endpoint security tooling.
Quantified measurement of control loss/recovery, detection accuracy, and legitimate-traffic impact within a controlled testbed.	Production deployment on a live commercial ISP network, or measurement of real subscriber traffic.

1.10 Limitations of the Study

The experimental findings are constrained by the composition of the controlled testbed and its traffic generators, which approximate but cannot fully replicate the diversity of real-world subscriber traffic. Detection performance may vary with the resolver software, blockchain-DNS client, and encrypted-transport implementation used by a given adversary or privacy-seeking user, and adversaries may adapt their traffic patterns in response to published detection techniques. The study measures technical control loss and recovery; it does not evaluate legal enforceability, subscriber consent, or the proportionality of any specific mitigation policy under a particular jurisdiction's law.

1.11 Definition of Key Terms

Term	Operational Definition in This Study
Decentralized Dynamic DNS (DDNS)	A naming and address-update mechanism in which name records are anchored on a distributed ledger or peer-to-peer system rather than a centrally operated registry, and are updated directly by the endpoint operator as its address changes.

Term	Operational Definition in This Study
ISP-level control	The set of technical capabilities, content filtering, lawful interception, malicious-domain blocking, and abuse tracking, that a network operator exercises by virtue of its position in the DNS resolution path.
Blockchain-anchored naming	A naming system, such as Namecoin, Handshake, or ENS, in which ownership and resolution data for a name are recorded on a blockchain and validated by network consensus rather than a central authority.
Encrypted DNS transport	A protocol, such as DoH, DoT, or DoQ, that carries DNS queries and responses inside an encrypted channel, preventing passive on-path observation of the query content.
Control-loss index	A quantified metric, defined in this study, expressing the proportion of naming/resolution activity in a traffic sample that is not observable or actionable through conventional ISP-level DNS controls.
Protective DNS	A managed DNS service, typically using response-policy zones or similar mechanisms, through which an operator blocks resolution of known-malicious domains as a security control.

1.12 Organization of the Thesis

Chapter	Proposed Content
Chapter 1	Introduction, research problem, scientific gap, purpose statement, research questions, objectives, significance, scope, limitations, and key terms.
Chapter 2	Literature review and theoretical background covering search logic, traditional DNS control, blockchain naming, encrypted transports, malicious abuse, detection research, and standards analysis.
Chapter 3	Research methodology and experimental design, including testbed architecture, traffic-generation rules, independent/dependent variables, metrics, and the evaluation protocol.
Chapter 4	Technical implementation details of the detection pipeline, feature engineering, classifier training, and mitigation policy engine.
Chapter 5	Experimental results, detection performance, control-loss/recovery measurements, and comparative baseline discussion.
Chapter 6	Research summary, operational recommendations, identified limitations, and future research directions.

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL BACKGROUND

2.1 Introduction

This chapter synthesizes the foundational concepts and prior research that inform this study. Rather than presenting isolated summaries, it evaluates how past research addresses traditional DNS governance, blockchain-anchored naming, encrypted DNS transports, malicious abuse of decentralized naming, and detection of encrypted/decentralized DNS traffic. This synthesis establishes the technical justification for the proposed detection and control-recovery framework.

2.2 Literature Search Logic and Screening Criteria

A component-based literature matrix was used across IEEE Xplore, ACM Digital Library, arXiv, MDPI, and official vendor, standards, and governance sources (CISA/NSA Protective DNS guidance, IETF RFCs for DoH/DoT/DoQ). Because the proposed study deliberately integrates several research strands, the review searches by component and then synthesizes the gap.

Search Block	Purpose in This Research	Representative Keywords
Blockchain-anchored naming	Establish the technical design and decentralization properties of alternative DNS systems.	Blockchain DNS, Namecoin, Handshake, Ethereum Name Service, decentralized domain name system
Encrypted DNS transports	Characterize how DoH/DoT/DoQ remove ISP-level plaintext visibility.	DNS over HTTPS, DNS over TLS, DNS over QUIC, ISP visibility, encrypted DNS filtering bypass
Malicious abuse of decentralized naming	Establish that decentralized/encrypted naming is already used for resilient malware C2.	Blockchain C2, decentralized command and control, DGA, malware naming resilience
Detection and classification	Inform the design of the proposed network-edge classifier.	DoH traffic classification, encrypted traffic fingerprinting, machine learning DNS detection, CIRA-CIC-DoHBrw
Standards and governance	Map the artifact and evaluation to recognized protective-DNS and encrypted-DNS guidance.	Protective DNS, NIST, CISA, IETF RFC 8484, RFC 7858

The screening principle classifies sources by closeness. Core sources directly address blockchain DNS, encrypted DNS visibility, or ML-based DNS traffic detection. Strong supporting sources are protocol-general (TLS/QUIC fingerprinting, malware C2 resilience) but provide reusable detection methodology or threat framing. Contextual sources, such as general privacy-technology overviews, are used only for background. Sources limited to unrelated topics are excluded.

2.3 Traditional DNS and the Basis of ISP-Level Control

Traditional DNS resolution places a subscriber's queries through resolvers the ISP operates or can observe, which is the technical basis for content filtering, parental controls, lawful interception, and abuse tracking of dynamically assigned addresses. This centralized, plaintext architecture is also the basis for Protective DNS services, in which an operator blocks resolution of known-malicious domains using response-policy zones or equivalent mechanisms. The same centralization, however, creates single points of failure and censorship susceptibility, which is the motivating concern behind decentralized alternatives.

2.4 Blockchain-Anchored and Peer-to-Peer Naming Systems

Namecoin was among the first systems to anchor domain ownership on a blockchain, but subsequent analysis identifies slow block times, limited throughput, and reliance on merge-mining with Bitcoin as centralization risks. Handshake decentralizes specifically the DNS root zone rather than all of DNS, replacing ICANN's control of top-level domain allocation with a proof-of-work blockchain while remaining backward-compatible with existing DNS resolution and DNSSEC validation; feasibility studies note that Handshake names require special client configuration and introduce reputational and technical risks for ordinary users. The Ethereum Name Service (ENS) and related projects extend blockchain-anchored naming to smart-contract platforms. Recent comparative surveys evaluate representative blockchain-DNS (BDNS) projects, including Namecoin, Handshake, and ENS, against record-model, finality, resolver-path friction, and interoperability criteria, and newer proof-of-work BDNS designs combine blockchain anchoring with IPFS-based distributed storage to reduce record-propagation latency.

2.5 Encrypted DNS Transports and the Erosion of Visibility

DNS-over-HTTPS (DoH) wraps DNS queries inside HTTPS, which blends DNS lookups into ordinary encrypted web traffic and removes the plaintext domain string from ISP-level observation; industry analysis identifies this loss of visibility, rather than performance, as the primary source of ISP objection to default DoH deployment, since visibility of DNS traffic is what allows many ISPs to perform lawful interception and content filtering. Enterprise-security literature confirms the same effect at smaller scale: DoH is documented to bypass DNS-based web filters and Protective DNS controls, and security guidance describes DoH as a channel already used by adversaries for command-and-control and data exfiltration specifically because it evades DNS-based visibility and filtering.

2.6 Abuse of Decentralized and Encrypted Naming by Malicious Actors

Malware-defense research directly documents the challenge that blockchain-based naming poses to network defenders: unlike a conventional domain, a blockchain-anchored C2 name cannot be seized, sinkholed, or taken down by a registrar, and defenders are left with only a small number of intervention points, most of which carry collateral-damage trade-offs similar to those of blocking Tor traffic outright. This body of work establishes that the threat this proposal investigates is not hypothetical: decentralized and encrypted naming channels are already part of the practical malware C2 landscape, which strengthens the case for a detection-and-mitigation response that is more targeted than blanket protocol blocking.

2.7 Detection and Classification of Encrypted/Decentralized DNS Traffic

A substantial body of work applies machine learning to encrypted-DNS traffic classification, most commonly using the CIRA-CIC-DoHBrw-2020 dataset and classifiers such as Random Forest, Gradient Boosting, AdaBoost, and LSTM-based deep learning, achieving reported accuracies in the low-to-high 90th percentile for distinguishing DoH from non-DoH traffic and, in some studies, benign from malicious DoH tunnels. Passive-analysis and fingerprinting studies further show that statistical features of encrypted DNS flows, timing, packet size, and burst structure, carry enough signal for classification even without decrypting payloads. This literature provides the methodological foundation for the network-edge classifier proposed in this study, though existing work targets encrypted DNS in general rather than decentralized-DDNS resolution and update traffic specifically.

2.8 Standards and Governance Blueprints

Governance guidance from CISA, the NSA, and equivalent national bodies promotes Protective DNS as a baseline security control, while IETF standards (RFC 8484 for DoH, RFC 7858 for DoT) define the encrypted transports that complicate that control's deployment. This tension, between encrypted-DNS privacy benefits and Protective-DNS security benefits, is acknowledged in the encrypted-DNS literature itself, which typically proposes managed-resolver policies or decryption/inspection proxies as a compromise. This proposal treats that guidance as the compliance and design target for the mitigation-policy layer evaluated in Chapter 3.

2.9 Comparative Review of Previous Studies and Technical References

The comparative review is organized into three focused groups. The first group supports the blockchain-DNS and decentralized-naming threat model. The second group supports the malicious-abuse and encrypted-visibility framing. The third group supports the detection and classification methodology.

A. Blockchain-DNS and decentralized-naming sources

Study / Source	Key Value	Limitation and Relevance to This Research
Yang et al., 2025 (arXiv 2508.05655)	Presents a proof-of-work blockchain-based DDNS integrated with IPFS, reporting sub-second query resolution and 15-second record propagation, and explicitly frames traditional DNS poisoning/censorship as the motivating threat.	System-design paper written from the builder's perspective; supports this study's characterization of decentralized-DDNS technical properties and propagation latency.
A Survey on Traditional DNS and Blockchain-Based DNS, MDPI 2026	Comparatively evaluates seven BDNS projects (including Namecoin, Handshake, ENS) on record model, finality, resolver-path friction, and DNSSEC/DoH-DoT interoperability.	Survey-level; motivates this study's selection of representative BDNS systems for testbed traffic generation.
Blockchain-based DNS: Current Solutions and Challenges to Adoption, CEUR 2024	Reviews BDNS solutions (Dot-P2P, Handshake, ENS, Namecoin, Blockstack, EmerDNS, Nebulis, OpenNIC) and adoption barriers.	Adoption-focused rather than security-focused; supports scoping of which BDNS systems are practically significant for a threat model.
Feasibility of Using Handshake Domain Names in Comparison with Classic DNS, 2024	Analyzes technical differences between Handshake and traditional DNS and their impact on security, censorship resistance, and usability.	Handshake-specific; supports detailed traffic-generation design for the Handshake component of the testbed.

B. Malicious abuse and encrypted-visibility sources

Study / Source	Key Value	Limitation and Relevance to This Research
The Challenges of Blockchain-Based Naming Systems for Malware Defenders, eCrime 2022	Directly documents blockchain-based C2 naming and analyzes the limited, collateral-damage-prone intervention points available to defenders.	Defender-challenge framing without a proposed detection architecture; this study builds directly on its threat model to design a targeted response.
ThousandEyes / industry analysis, “DNS-over-HTTPS Performance”	Documents that ISP objection to default DoH stems primarily from loss of visibility needed for lawful interception and content filtering, not performance.	Industry commentary rather than peer-reviewed research; used to ground the practical ISP-control framing in Chapter 1.
CMU SEI, “DNS Over HTTPS: 3 Strategies for Enterprise Security Monitoring”	Confirms that adversaries already use DoH for C2 and exfiltration specifically to evade DNS-based	Enterprise-scale guidance rather than ISP-scale; informs the mitigation-policy design in Chapter 3.

Study / Source	Key Value	Limitation and Relevance to This Research
	visibility and filtering, and outlines enterprise mitigation strategies.	
Security Boulevard, "Preventing DNS Filtering Bypass by Encrypted DNS", 2025	Frames the status-quo response (blocking DoH outright) as sacrificing DoH's privacy value, motivating more targeted approaches.	Practitioner commentary; supports the proportionality argument for the proposed detection-over-blocking approach.

C. Detection and classification sources

Study / Source	Key Value	Limitation and Relevance to This Research
Insights into DoH: Traffic Classification for DNS over HTTPS, ACM SNTA 2023	Demonstrates up to 98% classifier accuracy distinguishing DoH from other traffic using statistical flow features across multiple DoH providers.	Distinguishes DoH generically rather than decentralized-DDNS specifically; supplies the feature-engineering baseline for this study's classifier.
Detection of DoH Traffic Tunnels Using Deep Learning, MDPI 2023	Compares multiple ML algorithms (Random Forest, LSTM, SVM, and others) for detecting DoH tunneling, reporting strong accuracy with low false-positive rates.	Focused on tunneling/exfiltration rather than naming-system provenance; informs classifier-selection methodology.
A Lightweight Double-Stage Scheme to Identify Malicious DoH Traffic, PMC 2023	Uses the CIRA-CIC-DoHBrw-2020 dataset with hybrid ML methods to separate benign, non-DoH, and malicious DoH traffic.	Dataset and methodology directly reusable as a baseline for this study's benign/malicious/decentralized-DDNS three-way classification task.
CO-DEFEND: Continuous Decentralized Federated Learning for DoH Threat Detection, arXiv 2025	Proposes federated learning for DoH tunnel detection to address dataset scarcity and privacy constraints across multiple network operators.	Federated-learning scope exceeds this study's single-testbed design; motivates future extension of the proposed classifier to multi-operator settings.

2.10 Synthesized Research Gap

The reviewed literature establishes four facts. First, blockchain-anchored naming systems are mature enough to provide practical, low-latency, censorship-resistant name resolution, and are actively being extended (for example, with IPFS-backed storage) to improve performance and usability. Second, encrypted DNS transports demonstrably and intentionally remove ISP-level plaintext visibility, and this effect is treated by the ISP and enterprise-security communities as a genuine loss of control rather than a minor inconvenience. Third, malicious actors already exploit blockchain-anchored naming for takedown-resistant C2, and defenders report having few proportionate intervention options. Fourth, a mature methodology exists for machine-learning classification of encrypted DNS traffic, primarily targeting generic DoH detection rather than decentralized-DDNS resolution and update traffic specifically.

The refined research gap is therefore clear: existing work provides strong evidence for each component of the threat, blockchain naming, encrypted transport, malicious abuse, and generic detection methodology, but no study combines

them into a single, quantified account of ISP-level control loss, nor evaluates a detection-and-mitigation framework purpose-built for decentralized-DDNS traffic specifically, as distinct from encrypted DNS in general. This proposal occupies that gap directly.

2.11 Conceptual Framework of the Proposed Study

The conceptual framework links the observed gap to the proposed contribution. Existing ISP practice is treated as the baseline: legacy plaintext DNS filtering, do-nothing in the face of encrypted/decentralized traffic, or Protective DNS applied only to conventional domains. The proposed framework is expected to improve control recovery by combining network-edge flow fingerprinting, machine-learning classification tuned to distinguish decentralized-DDNS traffic from ordinary encrypted web traffic, a quantified control-loss index that makes the erosion measurable, and a targeted mitigation-policy layer that acts only on classified decentralized-DDNS flows rather than blocking encrypted DNS broadly. The framework will be evaluated using detection accuracy, false-positive impact, control-loss/recovery measurement, malicious-C2 blocking effectiveness, and performance overhead.

CHAPTER 3: RESEARCH METHODOLOGY AND EXPERIMENTAL DESIGN

3.1 Research Design

This study adopts a quantitative, design-science experimental methodology. The design-science component builds a network-edge detection pipeline, a quantified control-loss index, and a targeted mitigation-policy layer. The experimental component tests this framework against current ISP practice under controlled, representative traffic mixes of legitimate web browsing, privacy-seeking decentralized-DDNS use, and blockchain-anchored malicious C2 traffic, within a controlled network testbed.

3.2 Quantitative Nature of the Study

Methodological Element	Decision for This Study
Study family	Network-security-mechanism and traffic-classification evaluation rather than a human-centered survey.
Main evidence type	Labeled packet captures, classifier performance metrics, control-loss/recovery measurements, and inline-performance telemetry.
Comparison logic	Proposed detection-and-mitigation framework compared against ISP status-quo baselines under identical, labeled traffic mixes.
Measurement principle	Every metric is defined operationally before execution and computed from automated, repeatable capture-and-classification scripts.
Reproducibility goal	Testbed topology, traffic-generator configuration, dataset composition, classifier hyperparameters, and evaluation scripts will be documented.

3.3 Proposed Artifact and System Architecture

The proposed artifact sits at the network edge, between subscriber-side traffic and the upstream internet, and combines passive flow capture, feature extraction, classification, and policy enforcement.

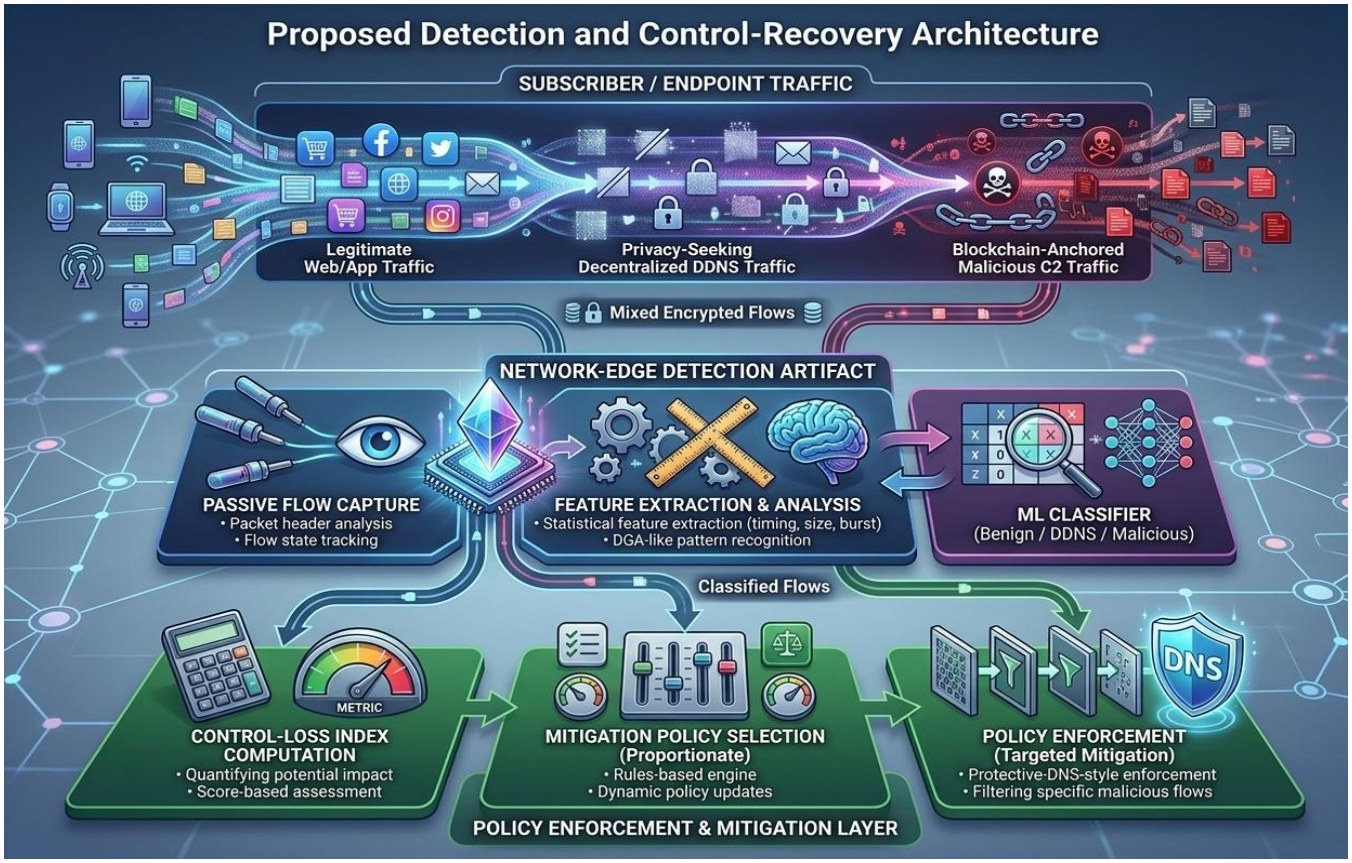


Figure 2: Proposed detection and control-recovery architecture.

3.4 Technical Design of the Detection and Mitigation Scheme

- Passive flow capture: mirrors traffic at the network edge without terminating TLS, preserving subscriber privacy for non-flagged flows.
- Feature extraction: derives statistical features (packet size distribution, inter-arrival timing, burst structure, TLS/QUIC handshake metadata, destination-endpoint churn) from encrypted flows without payload decryption.
- Classification: applies a trained machine-learning classifier to label flows as ordinary encrypted web traffic, decentralized-DDNS resolution/update traffic, or blockchain-anchored malicious C2 traffic.
- Control-loss index: computes the proportion of naming-related activity in a traffic sample that is not observable/actionable through conventional DNS-based controls, both before and after classification.
- Mitigation policy: applies targeted, proportionate action, such as Protective-DNS-style blocking, alerting, or rate limiting, only to flows classified as malicious, avoiding blanket blocking of legitimate encrypted or decentralized-DDNS traffic.

3.5 Threat Model and Assumptions

Threat-Model Component	Definition in This Study
Assets	ISP-level visibility into naming/resolution activity, content-filtering and Protective-DNS effectiveness, and abuse-tracking capability for dynamically assigned addresses.
Actors	Privacy-seeking subscribers adopting decentralized DDNS for legitimate reasons, and malicious actors (malware/botnet operators) adopting the same mechanisms for takedown-resistant C2.

Threat-Model Component	Definition in This Study
Capabilities	Both actor types can use blockchain-anchored naming (Namecoin, Handshake, ENS-style resolution) and encrypted transports (DoH, DoT, DoQ) to resolve and dynamically update names without ISP visibility.
Security objective	Recover a measurable degree of ISP-level control, particularly malicious-C2 blocking, without materially degrading legitimate privacy-seeking traffic or requiring blanket protocol blocking.
Out of scope	Breaking blockchain consensus or cryptographic guarantees, decrypting TLS/QUIC payloads, and legal/regulatory enforceability of any specific mitigation policy.

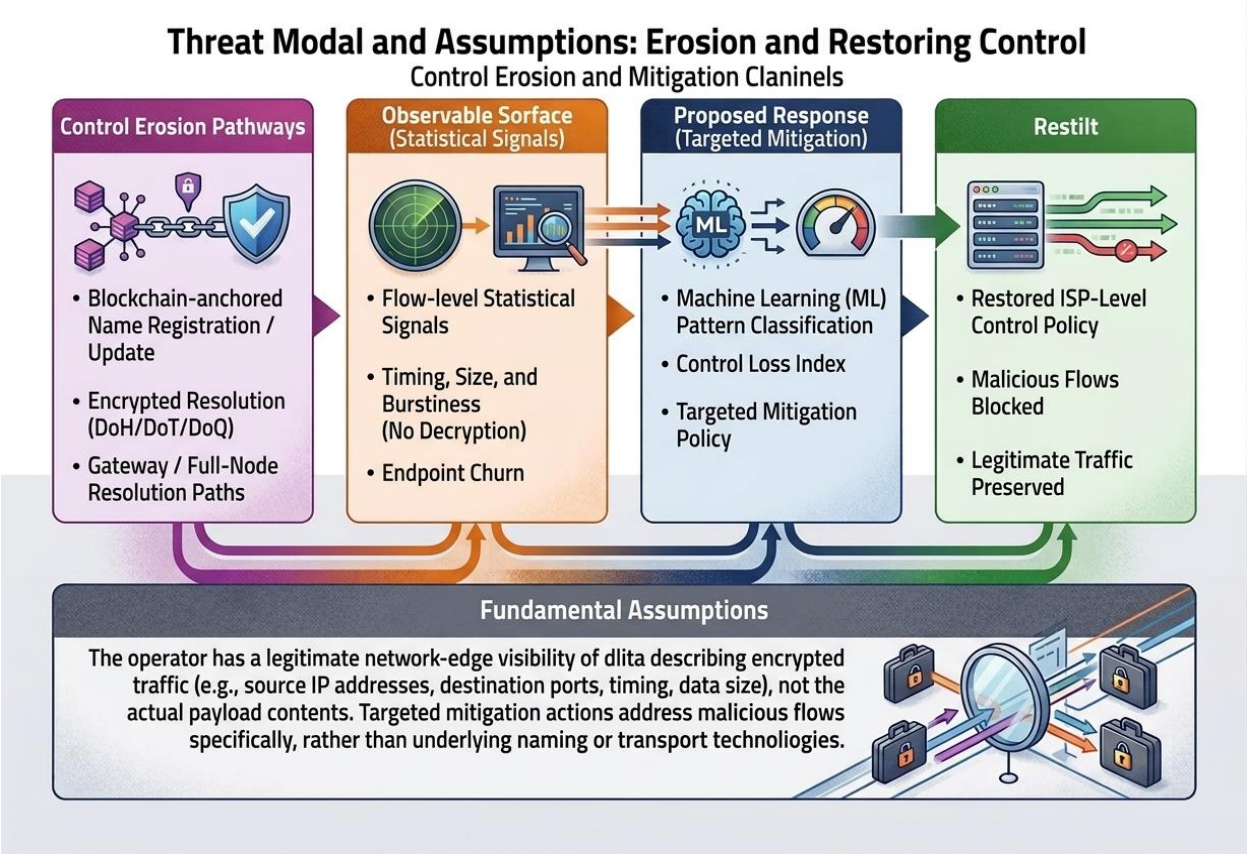


Figure 3: Threat model and control-erosion pathways.

3.6 Unit of Analysis and Observation

The primary unit of analysis is a labeled traffic flow, defined as a single resolution or update transaction (or an equivalent encrypted-session unit) generated under a defined condition: traffic category (benign web, legitimate decentralized-DDNS, malicious C2), naming/transport mechanism, and testbed configuration. Each unit records classifier output, ground-truth label, control-loss contribution, and any mitigation action taken.

3.7 Baseline Approaches and Planned Comparison

Defense Model	Visibility Retained	Vulnerability / Limitation	Expected Control Level	Evaluation Role
Do-nothing (status quo)	Only plaintext, non-decentralized, non-encrypted DNS traffic.	No visibility into DoH/DoT/DoQ or blockchain-anchored naming.	Low	Negative / control baseline.
Legacy DNS-based filtering	Plaintext DNS to ISP resolver only.	Fully bypassed by encrypted transports and decentralized naming.	Low–Medium	Legacy-practice reference.
Protective DNS (conventional domains only)	Known-malicious conventional domains.	Cannot enumerate or block blockchain-anchored names.	Medium	Security-control reference.
Proposed detection-and-mitigation framework	Classified encrypted and decentralized-DDNS flows via statistical fingerprinting.	Residual false negatives on novel/adapted traffic patterns; modest inline overhead.	High (targeted)	Primary method under study.

3.8 Variables and Metrics

Variable Type	Variables / Metrics
Independent variables	Detection/mitigation configuration, naming/transport mechanism mix, traffic volume, and testbed network conditions.
Dependent variables	Detection accuracy/F1, false-positive rate, control-loss index, malicious-C2 block success rate, inline latency/throughput overhead, and blocklist-maintenance churn.
Control variables	Testbed topology, traffic-generator configuration, dataset composition, and classifier training/test split.
Security metrics	True/false positive and negative rates for each traffic category, and control-loss index before/after mitigation.
Performance metrics	Inline processing latency, throughput impact, and CPU/memory overhead of the detection pipeline.
Usability metrics	Number of configuration/policy steps, alert volume, and estimated analyst review burden.

3.9 Research Question, Objective, and Metric Mapping

RQ	Aligned Objective	Primary Evaluation Metric
RQ1	Objective 1	Threat taxonomy and per-mechanism control-loss contribution.
RQ2	Objective 2	Comparative control-loss index across isolated vs. combined mechanisms.
RQ3	Objective 3	Classifier accuracy/F1 and false-positive rate on held-out traffic.

RQ	Aligned Objective	Primary Evaluation Metric
RQ4	Objective 4	Control-loss index before/after mitigation and malicious-C2 block success rate.
RQ5	Objective 5	Inline latency/throughput overhead and blocklist-maintenance churn.

3.10 Experimental Environment

Environment Element	Planned Specification
Testbed topology	A controlled, isolated network segment with a network-edge capture/classification point between simulated subscriber hosts and the upstream internet.
Naming/transport clients	Handshake full-node/resolver client, Namecoin (.bit) resolver, ENS-compatible gateway resolution, and DoH/DoT/DoQ-capable browsers and stub resolvers.
Traffic generation	Scripted generation of labeled benign web browsing, legitimate decentralized-DDNS resolution/update activity, and simulated blockchain-anchored C2 beaconing.
Classifier training data	A combination of the CIRA-CIC-DoHBrw-2020 dataset (as an encrypted-DNS baseline) and testbed-collected decentralized-DDNS and simulated-C2 captures.
Monitoring tools	Packet capture (tshark/Wireshark), flow-feature extraction scripts, and a Python-based ML pipeline (scikit-learn / gradient-boosting and LSTM baselines).

3.11 Traffic and Threat Scenarios

Scenario	Purpose	Expected Evidence
Benign encrypted web browsing	Establish the false-positive baseline for ordinary HTTPS/DoH traffic.	Classifier false-positive rate on labeled benign flows.
Legitimate decentralized-DDNS use	Test whether the classifier can identify privacy-seeking Handshake/Namecoin/ENS activity without over-flagging it as malicious.	Classification accuracy and category-level confusion matrix.
Blockchain-anchored simulated C2 beaconing	Test detection of periodic, low-volume beacon-like resolution/update traffic resembling malware C2 over decentralized naming.	Detection/block success rate and time-to-detection.
Mixed peak-load traffic	Evaluate detection accuracy and inline overhead when all traffic categories are combined at realistic volumes.	Latency/throughput impact and sustained classification accuracy under load.
Adaptive-adversary variant	Introduce traffic-pattern variation (timing jitter, padding) to test classifier robustness against evasion.	Accuracy degradation under adaptive conditions, informing limitations discussion.

3.12 Experimental Protocol

2. Configure the testbed with representative benign, legitimate decentralized-DDNS, and simulated malicious traffic generators.
3. Capture baseline traffic under the do-nothing and legacy-DNS-filtering configurations and compute the initial control-loss index.

4. Train and validate the detection classifier using the combined public and testbed-collected dataset, with a held-out test split.
5. Deploy the classifier and mitigation-policy layer inline within the testbed.
6. Rerun all traffic scenarios under the proposed framework and record classification outcomes and mitigation actions.
7. Record control-loss index, detection accuracy, false-positive impact, and inline performance metrics.
8. Repeat performance-sensitive runs 30 times to compute reliable descriptive statistics.
9. Compare outcomes across baselines and the proposed framework to determine control-recovery gains and operational cost.

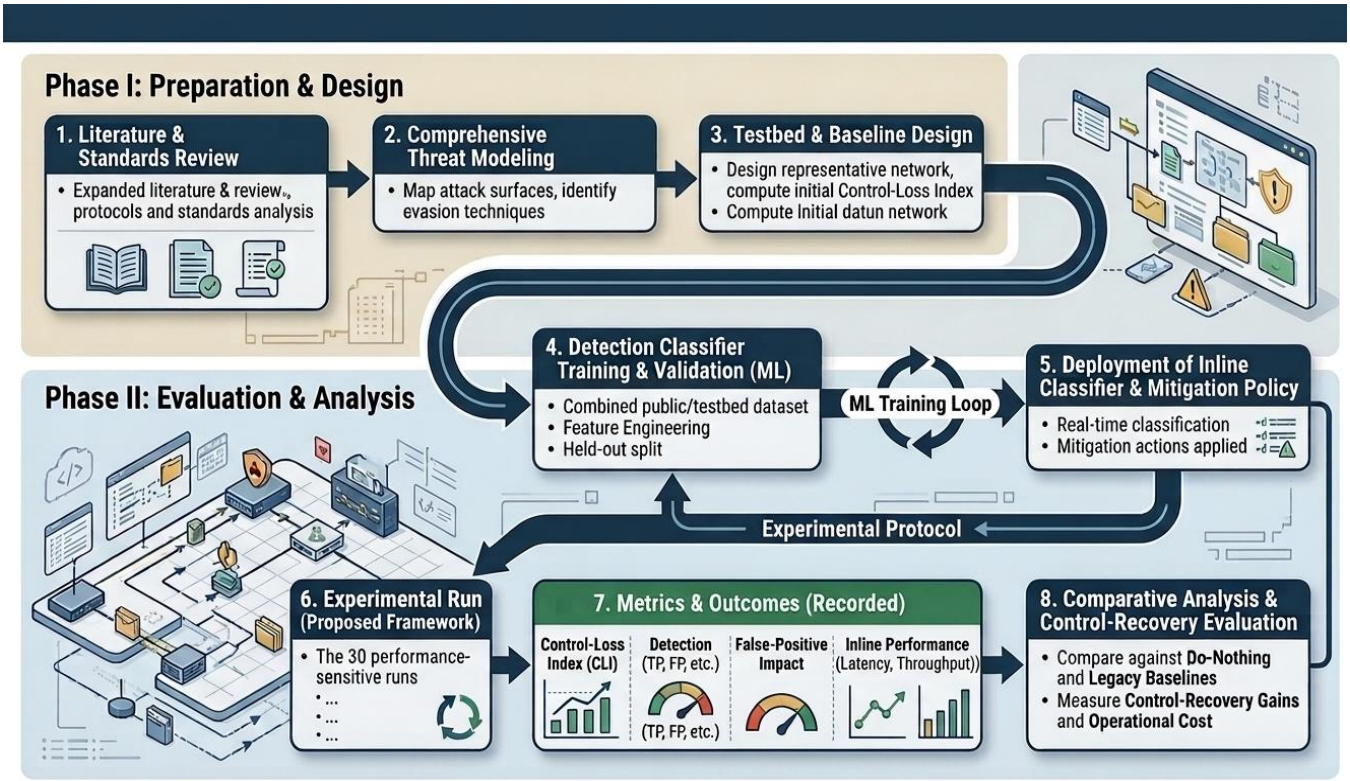


Figure 4: Proposed research methodology and evaluation pipeline.

3.13 Data Collection and Evidence Logging

Data collection relies entirely on synthetic and testbed-generated traffic together with the public CIRA-CIC-DoHBrw-2020 dataset; no real subscriber traffic or personal data is collected. Evidence includes labeled packet captures, classifier training/evaluation logs, control-loss index computations, and performance telemetry, indexed by scenario, run number, configuration, and timestamp.

3.14 Data Analysis Plan

Analysis Target	Planned Analysis
Detection performance	Compute accuracy, precision, recall, F1, and confusion matrices across traffic categories and classifiers.

Analysis Target	Planned Analysis
Control-loss/recovery	Compare the control-loss index across baseline and proposed configurations for each traffic scenario.
Performance overhead	Compute latency, throughput, and resource-utilization impact of inline classification and mitigation.
Robustness	Assess accuracy degradation under the adaptive-adversary scenario and discuss implications for deployment.
Standards alignment	Map the mitigation-policy design to Protective DNS and encrypted-DNS governance guidance.

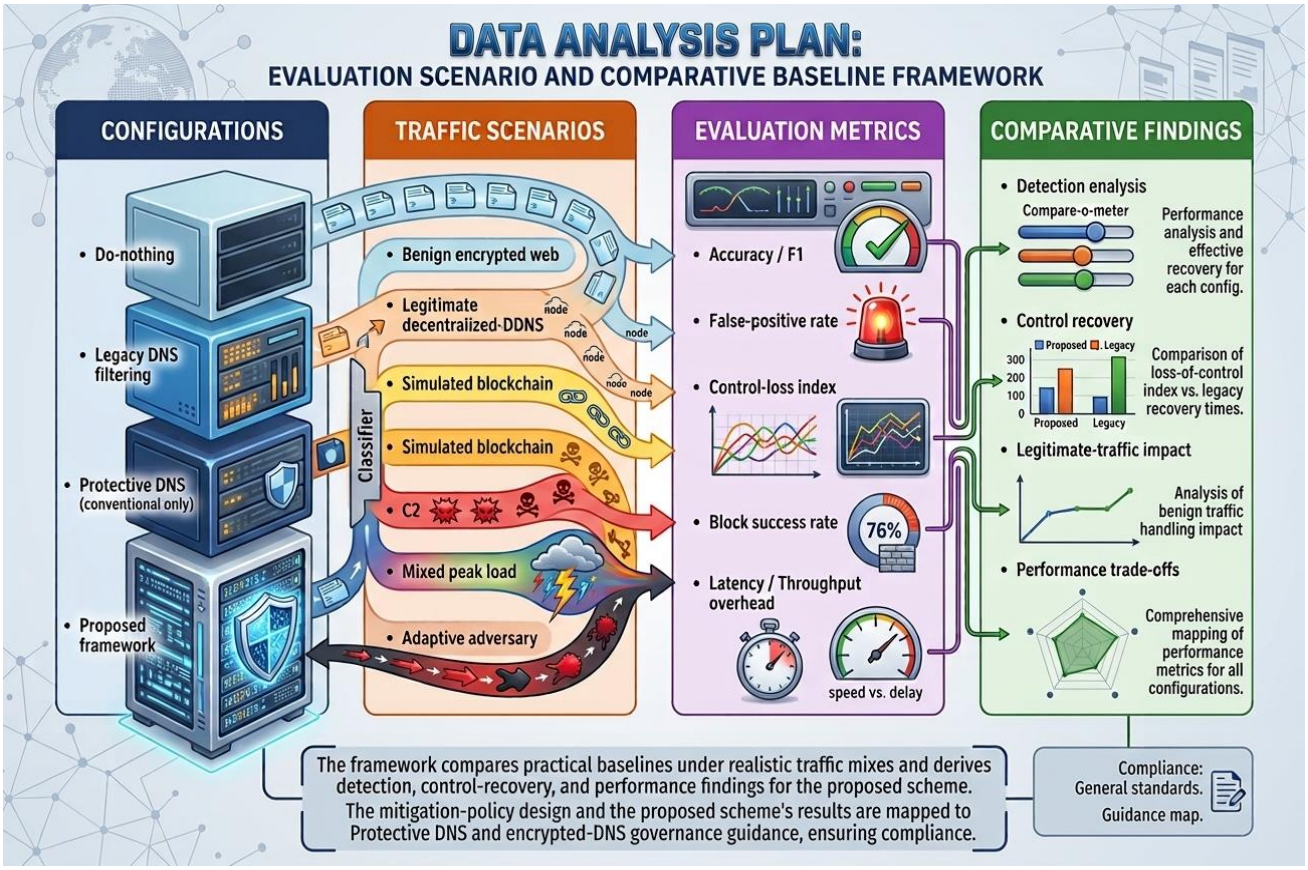


Figure 5: Evaluation scenario and comparative baseline framework.

3.15 Statistical Treatment and Repetition Rules

Measurement Item	Planned Statistical Treatment
Classifier evaluation	Use k-fold cross-validation on the training set and a held-out test split; report mean and standard deviation of accuracy/F1 across folds.
Warm-up and measured runs	Use 5 warm-up executions before performance measurement, then run each performance-sensitive scenario 30 times.

Measurement Item	Planned Statistical Treatment
Latency and resource metrics	Report mean, median, standard deviation, minimum, maximum, and 95% confidence intervals.
Non-normal or unstable measurements	Emphasize median and interquartile range where measurements are skewed, and report outliers transparently.
Control-loss/recovery outcomes	Report as percentages with confidence intervals across repeated traffic-mix runs.
Interpretation rule	Treat detection gains and false-positive/performance costs together: a configuration is useful only if it meaningfully reduces the control-loss index without disproportionate impact on legitimate traffic.

3.16 Validity, Reliability, and Reproducibility

Quality Concern	Mitigation Strategy
Internal validity	Use identical testbed configuration, traffic generators, and labeling procedures across all configuration variants.
Construct validity	Define each metric, including the control-loss index, operationally before execution.
External validity	Use realistic naming/transport clients and a public benchmark dataset to maximize relevance, while acknowledging limits to generalization to live ISP networks.
Reliability	Repeat classifier training/evaluation and performance measurements, and preserve all logs and trained models.
Reproducibility	Document testbed topology, software versions, dataset composition, classifier hyperparameters, and evaluation scripts.

3.17 Ethical Considerations

All testing is performed within an isolated, researcher-controlled testbed using synthetic and publicly available benchmark traffic; no real subscriber data, live ISP infrastructure, or production networks are used or targeted. Simulated malicious C2 traffic is generated and contained entirely within the testbed and is not directed at any external system. The study reports technical findings about detection and control recovery without making claims about the legality or policy desirability of any specific mitigation action.

3.18 Expected Results

Rather than claiming to eliminate the privacy or censorship-resistance benefits of decentralized DDNS, the study aims to demonstrate that a targeted, statistically grounded detection-and-mitigation framework can measurably reduce the control-loss index attributable to decentralized DDNS, and can meaningfully improve blocking of blockchain-anchored malicious C2 traffic, while keeping false-positive impact on legitimate encrypted and decentralized traffic, and inline performance overhead, within operationally acceptable bounds.

3.19 Proposed Timeline

Weeks	Main Tasks
1–4	Define threat model, control-loss index, testbed architecture, and metrics.

Weeks	Main Tasks
5–8	Complete the literature review and assemble the public and testbed-generated datasets.
9–12	Build the feature-extraction pipeline and train baseline classifiers.
13–16	Implement the mitigation-policy layer and integrate it inline within the testbed.
17–20	Execute all traffic scenarios, collect performance and control-loss metrics, and refine the classifier.
21–24	Analyze experimental data, complete standards-mapping documentation, and write the final thesis chapters.

PROPOSAL READINESS CHECKLIST

Status	Item	Evidence in Proposal
✓	Defined problem	Confined to network-edge detection of decentralized-DDNS traffic and its effect on ISP-level control.
✓	Testable research questions	Each RQ links directly to a specific measurement (control-loss index, classifier accuracy, performance overhead).
✓	Verifiable output artifact	Delivers a detection/classification pipeline, a quantified control-loss index, and a mitigation-policy template.
✓	Comparative testing baselines	Compares the proposed framework against do-nothing, legacy DNS filtering, and conventional-domain-only Protective DNS.
✓	Repeatable research steps	Outlines clear testbed parameters, traffic-generation scripts, and data-logging procedures.
✓	Stronger literature foundation	Chapter 2 uses component-based search blocks and prioritized sources across blockchain DNS, encrypted transports, malicious abuse, and ML detection.
✓	Quantitative treatment specified	Chapter 3 defines repetition rules, cross-validation, descriptive statistics, and baseline-comparison logic.
✓	Statistical foundations	Defines clear testing-repetition rules and data-validation guidelines (k-fold CV, 5 warm-up / 30 measured runs, 95% CI).
✓	Honest scope	The proposal avoids claims of eliminating decentralized DNS's privacy value and states realistic limitations.

CONCLUSION

This proposal outlines a structured, scientifically rigorous plan to characterize and quantify how decentralized Dynamic DNS erodes ISP-level network control, and to design, implement, and evaluate a targeted, proportionate detection-and-mitigation response. Rather than claiming to defeat decentralized naming or eliminate encrypted-DNS privacy benefits, the study focuses on making the resulting control loss measurable and on recovering a defensible share of that control specifically against malicious abuse.

The expected contribution is concrete: a reusable network-edge detection and classification artifact, a quantified control-loss index methodology, a targeted mitigation-policy template, and a comparative evaluation framework. The proposal is suitable for a master's thesis because it can produce both a working detection pipeline and measurable evidence, including classification accuracy, control-loss/recovery figures, and performance overhead, while remaining grounded in a substantial and rapidly growing body of literature on blockchain DNS, encrypted transports, and encrypted-traffic classification. Its main value is that it turns an often anecdotal concern, the erosion of ISP-level control by decentralized DDNS, into an integrated, testable, and quantifiable research and engineering problem.

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